

# Double power series with a radius margin

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## Abstract

The algebra of a coefficient array  $c : \mathbb{N}^2 \rightarrow \mathbb{R}$  with  $|c_{ij}|\rho^{i+j} \leq H$  on  $[0, b]^2$ ,  $b < \rho$  (Astra #6(a), route D): absolute summability of  $\sum c_{ij}u^i v^j$ , the faces  $a_i(v) = \sum_j c_{ij}v^j$  and  $b_j(u) = \sum_i c_{ij}u^i$ , the face remainders  $|a_i(v) - \sum_{j < M} c_{ij}v^j| \leq H\rho^{-i}(v/\rho)^M/(1-v/\rho)$ , the exact rectangular identity

$$\Phi - \sum_{i < M_1} u^i a_i - \sum_{j < M_2} v^j b_j + \sum_{i < M_1} \sum_{j < M_2} c_{ij} u^i v^j = \sum_{i,j} c_{M_1+i, M_2+j} u^{M_1+i} v^{M_2+j},$$

and the tail bound  $H\rho^{-M_1-M_2}u^{M_1}v^{M_2}/((1-u/\rho)(1-v/\rho))$ . Compatibility of the face jets is automatic because both faces are built from the same  $c_{ij}$ , so no symmetry of mixed partial derivatives is needed: this is the Schwarz-free route to the smooth/analytic  $d = 2$  theorem. Zero sorry/axiom.

## 1 The things formalised

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theorem summable_abs_of_geom (d :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (A r t :  $\mathbb{R}$ ) (hA : 0 < A) (hr0 : 0 < r)
  (hr1 : r < 1) (ht0 : 0 < t) (ht1 : t < 1)
  (hd :  $\forall ij, |d\ ij| \leq A * r^{ij.1} * t^{ij.2}$ ) :
  (Summable fun ij => |d\ ij|) < A / ((1 - r) * (1 - t))
theorem coeff_term_le ... :
  |c\ ij * u^{ij.1} * v^{ij.2}| < H * (u /  $\rho$ )^{ij.1} * (v /  $\rho$ )^{ij.2}
noncomputable def dblSum (c :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (u v :  $\mathbb{R}$ ) :  $\mathbb{R} :=$ 
  ' ij :  $\mathbb{N} \times \mathbb{N}$ , c\ ij * u^{ij.1} * v^{ij.2}
noncomputable def faceU (c :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (i :  $\mathbb{N}$ ) (v :  $\mathbb{R}$ ) :  $\mathbb{R} :=$  ' j :  $\mathbb{N}$ , c\ (i, j) * v^j
noncomputable def faceV (c :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (j :  $\mathbb{N}$ ) (u :  $\mathbb{R}$ ) :  $\mathbb{R} :=$  ' i :  $\mathbb{N}$ , c\ (i, j) * u^i
theorem dbl_summable_abs / dbl_summable / faceU_term_le / faceU_summable_abs
theorem faceU_rem_le (c :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (H v :  $\mathbb{R}$ ) (h : 0 < v)
  (hc :  $\forall ij, |c\ ij| * \rho^{-(ij.1 + ij.2)} \leq H$ ) (hv : 0 < v) (hv : v <  $\rho$ ) (i M :  $\mathbb{N}$ ) :
  |faceU c i v -  $\sum_{j < M} c\ (i, j) * v^j$ |
  < H * ( $\rho^{-i}$ ) * (v /  $\rho$ )^M / (1 - v /  $\rho$ )
theorem dbl_rect_identity (c :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (H u v :  $\mathbb{R}$ ) (h : 0 < v) (hH : 0 < H)
  (hc :  $\forall ij, |c\ ij| * \rho^{-(ij.1 + ij.2)} \leq H$ ) (hu : 0 < u) (hu : u <  $\rho$ ) (hv : 0 < v)
  (hv : v <  $\rho$ ) (M M :  $\mathbb{N}$ ) :
  dblSum c u v -  $\sum_{i < M} u^i * \sum_{j < M} c\ (i, j) * v^j$ 
  -  $\sum_{j < M} v^j * \sum_{i < M} c\ (i, j) * u^i$ 
  +  $\sum_{i < M} \sum_{j < M} c\ (i, j) * u^i * v^j$ 
  = ' ij :  $\mathbb{N} \times \mathbb{N}$ , c\ (M + ij.1, M + ij.2) * u^{M + ij.1} * v^{M + ij.2}
theorem dbl_tail_le ... :
  | ' ij :  $\mathbb{N} \times \mathbb{N}$ , c\ (M + ij.1, M + ij.2) * u^{M + ij.1} * v^{M + ij.2} |
  < H * ( $\rho^{-(M + M)}$ ) * (u^M * v^M) / ((1 - u /  $\rho$ ) * (1 - v /  $\rho$ ))

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## 2 Mathematics

Everything rests on the geometric double bound: a family dominated by  $Ar^i t^j$  with  $r, t < 1$  is absolutely summable with sum at most  $A/((1-r)(1-t))$  (product of two geometric series via `summable_mul_of_summable_norm` and `tsum_mul_tsum_of_summable_norm`). The rectangular identity is proved by writing the double sum as an iterated sum (`Summable.tsum_prod`), splitting each row at  $M_2$  and the resulting column sums at  $M_1$  (`Summable.sum_add_tsum_nat_add`), interchanging the finite sum over  $j < M_2$  with the  $i$ -sum (`Summable.tsum_finsetSum`), and identifying the doubly shifted tail with the product sum over the shifted index set (`Summable.comp_injective` for the shift, then `tsum_prod` again); the summability of the row tails in  $i$  comes from the summability of the row absolute sums (`summable_prod_of_nonneg`).

## 3 Paper-facing meaning

An amplitude given by a convergent double series with a radius margin, the standing situation for analytic  $\xi, \eta$  near the exceptional divisor, produces exactly the data of the  $d = 2$  cutoff theorem: compatible faces, face remainders and a mixed remainder with geometric constants, with the same  $s$ -envelope  $H(s)$  throughout. The next unit attaches the parameter  $s$ , supplies the continuity hypotheses (via a clamp of the coordinates to the box) and the `FaceMoments` through `faceMoments_of_envelope`, giving the analytic  $d = 2$  cutoff theorem; the coefficient closure for  $\eta e^{\beta s \xi}$  (Astra #6 rank 2) then discharges the series hypothesis.

## 4 Lean notes

- `Summable.comp_injective` names its injection `i`, not `f`.
- `pow_add` in `rw` hits every  $x^{m+n}$ , including inside an absolute value that must stay intact; pass the base (`pow_add (v / )`) to restrict it.
- Row-tail summability in the outer index follows from `Summable.of_norm_bounded` with the row absolute sums, which `summable_prod_of_nonneg` makes summable.
- `Summable.tsum_finsetSum` is the tsum-of-finite-sum lemma (not `tsum_sum`).
- After `tsum_geometric_of_lt_one`, close  $a \cdot (1 - q)^{-1} = a/(1 - q)$  with `ring`, not a targeted `div_eq_mul_inv` rewrite.